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Verification of on-demand medical device networks

Abstract: We propose a novel method for the automated formal analysis of on-demand medical device networks communicating via an open communication protocol. The resulting toolbox aims at assisting operators of cross-vendor medical device networks in the complex process of conformity assessment. The method is based on the notion of timed, extended finite automata and employs existent TCTL model checking techniques by use of MATLAB and UPPAAL. We evaluate our methodology in a real-life application similar to a user experience published by the U.S. Food and Drug Administration. The method revealed a flaw in the connector design for the Surgical Real Time Bus of the OR.Net project.

Keywords: Medical devices, admission process, networks, automata, verification.

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1 Medical device integration

With an ever increasing importance of digital data in the operating room [1], a trend towards manufacturer-independent medical device integration is in the rise [2]. The overall goals behind open medical device communication are [3, 4]

- efficient information aggregation
- increased procedure efficiency
- time and cost reduction and
- increased operation quality.

Typical applications for integrated medical devices include retrieval and archival of patient data in clinical

information systems, mirroring sensor data (e.g. patient pulse, oxygen saturation etc.) into medical live-images, automated control of devices (e.g. based on navigation data) and many more.

However, although technical requirements (bandwidth, latency, jitter) are met by various industrial communication standards, European regulatory guidelines present novel obstacles. In Germany, operators of interconnected medical devices must prove conformity of the network in question with the General Requirements [5,6], which in turn requires in-depth knowledge of every single network component. Hence, the mere technical possibility of cross-manufacturer data exchange does not imply its legal admissibility.

Our aim is to introduce a novel method for the automated verification, whether a set of formalized requirements can be met by a given medical device network. The resulting toolbox aims at shifting the complexity of conformity assessment from lay operators back to component manufacturers.

The rest of the article is organized as follows: In Section 2, we briefly depict the current situation regarding conformity assessment and point out the drawbacks of the state of the art. We then briefly sketch the description methodology in Section 3 – a detailed description with mathematical rigor is outside the scope of this paper. Section 4 presents a prototype implementation. In the end, we describe a first application in a real-life scenario and give an outlook for future developments.

2 State of research

A noticeable amount of international research projects addresses the issues of medical device interconnection. In the German flagship project OR.Net, a hybrid network architecture consisting of a webservice-based framework for non-time-critical data and a realtime framework for time-critical data was developed. The realtime framework *Surgical Real Time Bus* (SRTB), developed by the authors in [7, 8] makes use of the Ethernet Powerlink (EPL) communication

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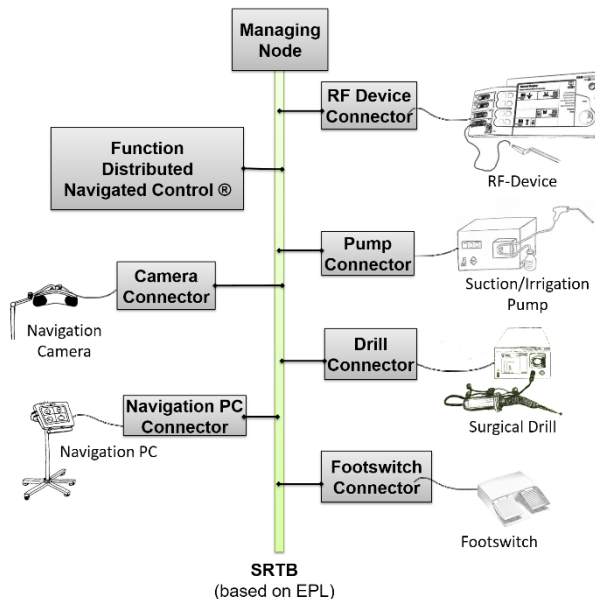


Figure 1: Simplified version of the SRTB. Medical devices and equipment are connected to the open communication bus via connectors. The Managing Node is responsible for the synchronization and data routing.

standard that guarantees realtime data transmission by use of the Time Division Multiple Access bus access scheme. It consists of one unique managing node and an arbitrary number of controlled nodes (max. 100). Medical devices communicate over the network via so-called connectors that act as translators between the proprietary medical device interfaces and the SRTB. Each connector is an EPL controlled node. The managing node synchronizes all controlled nodes and is responsible for the matching of data sources and sinks.

While much attention is paid to technical interoperability [9, 10], little is known about admissibility of cross-manufacturer device networks.

3 Approach

The task of this work is the conception and implementation of a software toolbox for the efficient, automated requirements verification of the SRTB. Once implemented, we expect the toolbox to aid lay users in the correct (less errors) and fast (shorter time) requirements verification of a given network. The core idea behind the novel methodology is to model every single network component as a system of communicating automata. More precisely, each controlled node is modelled as two concurrently executed automata – one representing the EPL-Process and one representing the actual device behavior. Moreover, the managing node will be described as a single automaton executed concurrently with all controlled nodes.



States Data Control Signals

Figure 2: Minimal requirements regarding the expressiveness of the description methodology. It must comprise states and state transitions, data and data dependencies as well as control signals.

Given a fixed data routing, the entire network is again described as a system of communicating automata that can be analyzed with model checking techniques. Before introducing the description method, it is worth determining the required expressiveness of the description methodology. Fig. 1 shows a suction/irrigation pump used in every endoscopic intervention (>1 Mio. occurrences p.a. in Germany). It has an On/Off button, various buttons for control signals and a display to show the current flowrate. Before operation, the pump must be initialized via a dedicated button. Once initialized, a simple on/off footswitch activates the pump, which in turn extracts liquid with the desired flowrate. This description reveals that the minimum set of elements needed to describe consists of

- States (e.g. on/off)
- State Transitions (e.g. off->on->initialized->active)
- Data (e.g. desired flowrate)
- Functional data dependencies (e.g. actual flowrate is a function of input signal and desired flowrate)
- Control Signals

In addition, the EPL communication technology requires

- Synchronization signals (start of cycle, poll request, poll response)
- Timing (cycle length, response times)

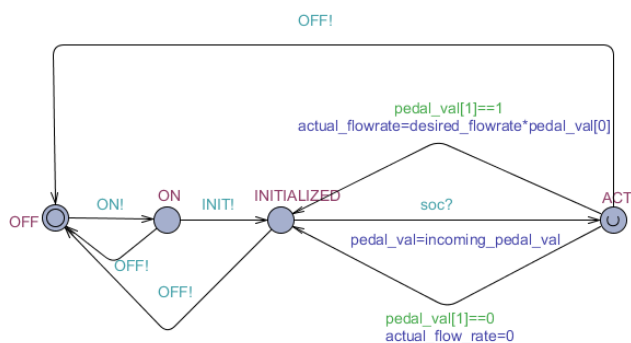


Figure 3: Image of a simplified timed, extended finite automaton representation of the suction/irrigation pump.

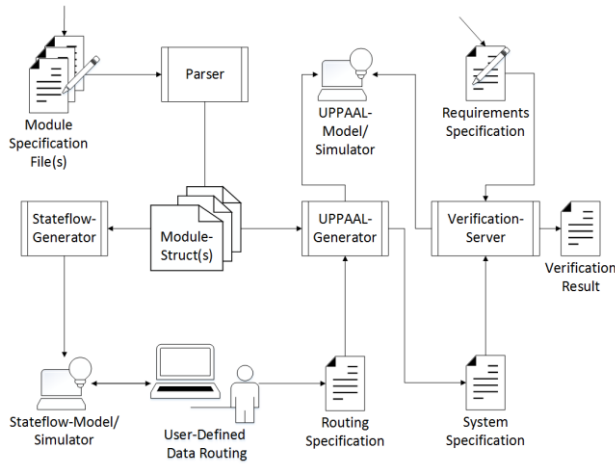


Figure 4: Overview over all components of the automated verification methodology.

Hence, the description methodology we chose are *Timed, Extended Finite Automata* (TEFA). While the formal introduction is out of the scope of this paper (cf [11]), a typical automaton of this class is depicted in Fig. 2. The figure shows a simplified version of a suction/irrigation pump. The semantics is as follows: The pump starts in the OFF state. Upon receipt of the ON! and INIT! signal, it changes to the INITIALIZED state. Once the input variable “incoming_pedal_val” is parsed and evaluated as valid (pedal_val[1]-bit), the pump realizes the desired flowrate.

Once each system component is described as above (by the manufacturer), the entire system can be analyzed by use of model-checking techniques.

Among all requirements specification languages, Timed Computation Tree Logic (TCTL) appears most apt for our applications.

4 Implementation

Exemplarily, we implemented the novel methodology with Matlab and UPPAAL [12]. The devices descriptions (edited by the respective device manufacturer) are simple ASCII text files, containing comma-separated arrays that represent the upper elements in a standardized fashion.

The device description files are read by Matlab and parsed into device structs (one per device). Once finished, the Stateflow Generator generates a Stateflow-Chart for each device. The user can then edit the signal routing (as in [13]). Once finished, the system specification is computed and written into a UPPAAL-readable XML-File.

The requirements specification is edited by both the manufacturers of a given device (technical requirements such as max. latency – part of the description file), as well as the

system operator (intervention-specific, clinical requirements – separate specification file) formulated as TCTL-formulae.

5 Example use-case

The U.S. Food and Drug Administration (FDA) publishes user experiences with faulty devices in their publically accessible MAUDE (Manufacturer and User Facility Device Experience) database. On the 6th of march, a professional user issued the following occurrence: A footswitch was used to activate a device delivering rf energy. After the surgeon released the footswitch pedal, the device continued to deliver rf energy for more than 3 seconds. Fortunately, no unintended tissue damage was experienced [14].

This type of issue is representative for a greater class of issues known as “real time requirements” that led to the development of a hybrid network architecture in the OR.Net project. Guarantees that safety-critical control signals arrive within predefined time boundaries are of vital importance in medical applications. We evaluated our approach in a similar usecase with a surgical suction/irrigation pump.

The requirement we layed down was: “Whenever the surgeon releases the footswitch, the pump will turn off within 50 ms”. Depending on the nomenclature, this may reduce to the formula

$$\forall \Box [(\text{Foot. OffTimer} \geq 50 \wedge \text{FootVal} = 0) \Rightarrow \text{flowrate} = 0]$$

As it turned out, the very first implementation of the SRTB did not satisfy the upper TCTL-formula, because it relied on the EPL connection to never fail. Hence, the pump connector was programmed to always use the last incoming pedal value transmitted via EPL to control the device. However, should the SRTB fail (e.g. due to a pulled out wire) while the pump was active, the pump would continue operation even if the surgeon stepped down from the pedal.

The sequence of steps to induce a requirement violation hence is:

1. step on the footswitch
2. the pump is activated
3. interrupt the connection
4. step down from the pedal and leave connection interrupted

A simple re-design of the pump connector solved the issue: A timer now watches over the EPL stack. Should the time span between two incoming control values exceed 15 ms, the pump will automatically turn off. Re-verification of the novel connector design now proves that the requirement can be met even in the event of network failure.

6 Conclusion and outlook

We introduced a toolbox that allows fast verification of on-demand medical device networks. The system aims at allowing a faster and more correct risk assessment of medical device networks by lay users. By using our method, part of the Know-How required to assess hazardous situations is transferred back from the operator to the manufacturer. The required description method is based on timed, extended finite automata. Compliant description files are simple ASCII text files and by parsing the files, a model of the entire system and its interactions can be generated automatically. The user can simulate the resulting system and formally verify a given set of requirements that are formulated as TCTL formulae. However, while the toolbox may be helpful in detecting hazardous situations, the overall risk assessment remains a challenging task. First, because not all requirements can be formulated as TCTL formulae ([15]) and second, because even if the toolbox directly pointed at possibly mitigating measures, the lay user will not be able to implement any of them himself. Hence, further risk management methods are required.

Author's Statement

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